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Teat preparation system

Abstract

A teat preparation apparatus to clean bovine teat before milking and then sanitize the teat after milking. The apparatus, which is connected to a control unit, includes a holder and a teat receiving body with high-pressure air, pre-dip, and post-dip ports. Inside the teat receiving body is a spray cylinder that forms two or three stacked, isolated raceways that communicate with the three ports so that air and the two solutions can be selectively-sprayed individually or together over a teat at different stages of operation. The holder may be attached a support arm integrated with a robotic system with a track parallel to a row of bovines to be milked. When used in a robotic system, two or more teat receiving bodies may be adjusted, enabling the spacing and orientation of each teat receiving body to be adjusted. Additionally, a deflection cap with an optional diffuser enhances functionality.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6302058	12/2000	Dahl	119/14.47	A01J 7/04
10945406	12/2020	Neal	N/A	A01J 7/04
2019/0133069	12/2018	Stuessel	N/A	A01J 7/022

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2017071535	12/2016	WO	A01J 7/04

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Background/Summary

REFERENCES TO RELATED APPLICATIONS (1) This continuation-in-part patent application is based on and claims the priority date benefit of the U.S. continuation-in-part patent application (application Ser. No. 17/324,984) filed on May 19, 2021, which is based on and claims the filing date benefit of the U.S. utility patent application (application Ser. No. 17/009,561) filed on Sep. 1, 2020, which is based on and claims the filing date benefit of U.S. provisional patent application 62/960,875 filed on Jan. 14, 2020.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention pertains to devices that automatically clean, dry, and apply a post-dip solution to individual teats on a bovine.

2. Description of the Related Art

(2) Dairy workers should follow good, hygienic milking procedures to ensure the milk collected is safe to consume. For example, each teat should be thoroughly cleaned before milking. The cleaning task is performed by a worker who stands on one side or behind the cow and applies a cleaning solution. The pre-dip solution is usually an aqueous mixture containing 0.5% iodine, hydrogen peroxide, chlorine dioxide, or a combination. After all teats have been cleaned with the pre-dip

solution, the worker uses a dry towel to remove the pre-dip solution and any remaining dirt or debris. Hundreds of towels are used daily, a substantial burden to dairy operators.

(3) Before milking, all teats are usually 'stripped' to stimulate oxytocin release. While washing and drying the teat may partially stimulate oxytocin release, the worker will sometimes manually squeeze and pull each teat downward to stimulate oxytocin release further. The amount of stimulation is dependent on the worker's skill and the amount of time spent manually squeezing and pulling the teats.

(4) After milking, the worker manually applies a disinfecting solution, called a 'post-dip solution', to each teat. The post-dip solution contains a higher concentration of iodine and a thickening agent, making it more viscous than the pre-dip solution. The post-dip solution must be applied to the tip of each teat and given sufficient time to dry in place and form a protective layer.

(5) Suppose the steps of applying and drying the pre-dip solution, stripping the teat, applying the post-dip solution, and allowing the post-dip solution to dry to form a protective layer are not completed. In that case, milk from the cow may be contaminated, and infection may occur.

(6) It is well known to dairy farmers that the size and shapes of teats on a cow vary. It is also well known to dairy farmers that the amount of dried mud on a bovine's teats also varies. Variations in the sizes and shapes of the teats and the amount of dried mud on the teats make consistently cleaning the teats on all cows in a dairy farm challenging.

(7) Because the worker must stand to one side or behind the cow to reach all of the teats, areas of the teats opposite the worker are challenging to reach. As a result, some areas on the teat are not treated with a pre-dip solution, dried, stripped, or treated with a post-dip solution.

(8) What is needed is a teat preparation system that uses a single, lightweight handpiece that thoroughly cleans, evenly applies a pre-dip solution to each teat, removes excessive pre-dip solution from each teat, and eliminates the use of cloth or paper towels.

(9) What is also needed is a teat preparation system that evenly applies a post-dip solution to the entire teat.

(10) What is also needed is a teat preparation system that can be easily modified to strip the teat.

(11) What is also needed is a teat preparation system that uses exchangeable components that enable different sizes and shapes of teats to be cleaned, sprayed with a pre-dip solution, dried, and sprayed with a post-dip solution.

(12) What is also needed is a teat preparation system that allows the user to selectively adjust the operation of the handpiece to clean the teat when needed.

SUMMARY OF THE INVENTION

(13) A teat preparation system automatically and selectively cleans the teats individually on a bovine with high-pressure air and a pre-dip solution that dries each teat without using towels after applying the pre-dip solution. The system then applies a post-dip solution to each teat after milking.

(14) The system includes a holder connected to a control unit. In one embodiment, the holder is a compact, lightweight handpiece, and configured to be held and operated with one hand. The handpiece is connected to the control unit, made with durable components that stand up to rough use in a milking parlor. In another embodiment, the holder is a support arm in a robotic control system. In both embodiments, the control unit controls high-pressure air delivery, a pre-dip solution, and a post-dip solution to a teat receiving body. The holder positions the teat receiving body under a teat during use, and the control unit is activated. When the holder is a handpiece, a mode switch and a trigger are provided that allows the user to manually control the delivery of the high-pressure air, the pre-dip solution, and the post-dip solution from the control unit to the handpiece.

(15) The teat receiving body includes a spray cylinder. A central bore configured to receive a teat is formed inside the spray cylinder. The central bore has a narrow top opening configured to receive and fit around a teat and a wide bottom opening through which high-pressure air and the pre-dip solution are discharged. The spray cylinder includes a wide upper section below the teat receiving

body's top opening in which two or more nozzles are formed. The nozzles are arranged in a circular pattern on the inside surface of the spray cylinder. The spray cylinder is replaceable and exchangeable with different spray cylinders, central bores with different diameters, and two to three circular raceways. The sidewalls of the spray cylinders are configured to form two or more isolated circular raceways between the inside surface of the teat receiving body and the outside surface of the spray cylinder. The two or more sets of nozzles are located on the inside surfaces of the central bore adjacent to the raceways so that air, pre-dip solution or post-dip solution delivered to the raceways flow through the set of nozzles adjacent to the raceway.

(16) The lower sidewalls of the spray cylinder below the lowest raceway diverge, creating a venturi when high-pressure air flows downward in the spray cylinder.

(17) Loaded into the control unit's memory is a software program that controls the delivery of high-pressure air, pre-dip solution, and post-dip solution to the holder. The software program controls the sequence and release of high-pressure air, pre-dip solution and post-dip solution, and the time each is delivered to the teat receiving body. For example, after the pre-dip solution has been delivered to the teat receiving body for a predetermined amount of time and discontinued, the control unit continues to deliver high-pressure air to the teat receiving body to remove the excessive pre-dip solution from the teat. After a suitable amount of time, the high-pressure air delivery is discontinued. The teat receiving body is then removed from the teat.

(18) After milking, the holder repositions the teat receiving body under the teat, and high-pressure air is selectively delivered. As mentioned above, a venturi is created in the spray cylinder that draws the teat into bottom opening in the spray cylinder. Once the teat is drawn into the bottom opening, the post-dip solution is then delivered to the handpiece. The high-pressure air is applied simultaneously with the post-dip solution to atomize the post-dip solution. After a predetermined amount of post-dip solution has been applied, the high-pressurized air is discontinued, and the application of the post-dip solution continues for a few seconds. The post-dip solution is then air-dried, forming a wet layer of post-dip solution over the entire teat. The teat receiving body is removed, and the post-dip solution can air dry.

(19) In another embodiment, the holder may be a support arm integrated with a robotic arm control apparatus connected to the control unit. When the holder is a handpiece, the handpiece includes a set of internal conduits that connect to external conduits connected to storage containers that hold the high-pressured air, the pre-dip solution, and the post-dip solution. When the holder is integrated into a robotic arm control apparatus or when a holder is a handpiece, each storage container is coupled to a solenoid that connects to the control unit. The control unit includes a programmable logic controller (also called a PLC). A software program controls the order in which the solenoids are opened and how long each solenoid is opened and closed. The software program is programmable, enabling the operator to adjust when and how long the solenoids are activated.

(20) The distal ends of the internal conduits in the handpiece or a support arm on a robotic unit connect to ports formed on the teat receiving body. In one embodiment, the distal ends of the conduits and the ports are horizontally aligned. In another embodiment, the distal ends of the conduits and the ports on the teat receiving body handpiece are vertically aligned, enabling an exchangeable teat-receiving body to be used. In this embodiment, the exchangeable teat receiving body is modified and configured to be used with a modified spray cylinder, respectively. The modified spray cylinder can be manufactured in different sizes, enabling the handpiece to be used with different teats.

(21) In still another embodiment, a deflection cap may be integrally formed or attached to the lower edge of the teat receiving body or to the spray cylinder. The deflection cap curves downward and inward to a narrow apex. During use, the deflection cap redirects the high-pressure air, the pre-dip solution, and the post-dip solution inward to form a narrow stream. The user may manipulate the handpiece to redirect the high-pressure air stream, pre-dip solution, and post-dip solution in dirty, hard-to-reach areas. An optional diffuser and agitator may be used inside the deflection cap to

enhance the distribution of the pre and post-dip solutions.

(22) In another embodiment, the teat receiving body is modified to include an elastic bladder sleeve inside a rigid frame. The bladder frame fits into a modified spray cylinder configured for use in the teat receiving body. The bladder sleeve is configured to squeeze against the outside surface of a teat placed inside the teat receiving body. The bladder sleeve squeezes downward, milking-like, against the sides of the teat to strip the teat after the pre-dip solution application and the drying steps have been completed.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates the teat preparation system disclosed herein, which includes a handpiece connected to a control unit.
- (2) FIG. 2 is a side elevational view of the handpiece.
- (3) FIG. 3 is a top plan view of the handpiece.
- (4) FIG. 4 is a bottom plan view of the handpiece.
- (5) FIG. 5 is a front elevational view of the handpiece.
- (6) FIG. 6 is a sectional, side elevational view of the handpiece.
- (7) FIG. 7 is a sectional, partially exploded, rear perspective view of the handpiece.
- (8) FIG. 8 is a sectional, partially exploded, front perspective view of the handpiece.
- (9) FIG. 9 is a rear elevational view of the spray cylinder.
- (10) FIG. 10 is a sectional side elevational view of the spray cylinder shown in FIG. 9.
- (11) FIG. 11 is a top plan view of the spray cylinder.
- (12) FIG. 12 is a perspective view of another embodiment of the handpiece.
- (13) FIG. 13 is a left-side elevational view of the handpiece shown in FIG. 12.
- (14) FIG. 14 is an exploded perspective view of the handpiece shown in FIG. 12.
- (15) FIG. 15 is a side elevational view of one-half of the handpiece shown in FIG. 12.
- (16) FIG. 16 is an end sectional view of the handpiece taken along line 16-16 in FIG. 13.
- (17) FIG. 17 is an end, sectional view of the spray cylinder taken along line 17-17 in FIG. 13.
- (18) FIG. 18 is a sectional, side elevational view of the spray cylinder used in the handpiece shown in FIGS. 12-17 that has an upper raceway that communicates with the air port and a lower raceway that communicates with the lower raceway.
- (19) FIG. 19 is a sectional, side elevational view of another embodiment of the spray cylinder that has an upper raceway, an intermediate raceway and a lower raceway that communicate with the air port, the pre-dip solution port, and the post-dip solution port, respectively.
- (20) FIG. 20 is a sectional, side elevational view of another embodiment of the spray cylinder with smaller diameter central port than the spray cylinder shown in FIG. 19 and has an upper raceway that communicates with an air port and a lower raceway that communicates with a pre-dip solution port or a post-dip solution port.
- (21) FIG. 21 is a perspective view of the handpiece with a spray cylinder with the deflection cap attached to the end of the teat receiving body, located inside the deflection cap is a diffuser with an agitator.
- (22) FIG. 22 is a side elevational view of the teat receiving body with the deflection cap, diffuser, and agitator.
- (23) FIGS. 23, 24, and 25 are sequential, side elevational views of a stripping embodiment of the invention, including an expandable, elastic bladder sleeve.
- (25) FIG. 26 is a front perspective view of the bladder frame used in the embodiment shown in FIGS. 23-25.

- (26) FIG. **27** shows a sectional side elevational view of the bladder frame and an elastic sleeve placed therein.
- (27) FIG. **28** is a perspective view of the lower support frame.
- (28) FIG. **29** shows the interconnections of the components used in the non-stripping embodiment shown in FIGS. **1-22**.
- (29) FIG. **30** is a diagram showing the interconnections of the components used in the system that uses the stripping embodiment shown in FIGS. **23-28**.
- (30) FIG. **31** is a perspective view of a robotic head assembly with one teat receiving body attached to a single rotating support arm.
- (31) FIG. **32** is a perspective view of a robotic head assembly with two teat receiving bodies attached to a single support arm.
- (32) FIG. **33** is a perspective view of a robotic head assembly with four teat receiving bodies attached to a single support arm.
- (33) FIG. **34** is a partial perspective of a robotic head assembly that uses four teat receiving bodies attached to two moveable mounting arms that are perpendicularly aligned to an alignment arm attached to the single support arm and moved to a narrow configuration n.
- (34) FIG. **35** is a partial perspective of a robotic head assembly similar to the robotic head assembly shown in FIG. **34** showing the mounting arms in a wide configuration.
- (35) FIG. **36** is a perspective view of the robotic multiple head assembly mounted on a track that enables the head assembly to move longitudinally over the track.

DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

- (36) FIG. **1** shows a first embodiment of a teat preparation system **8** used to automatically and selectively clean teat **5** on a bovine with a pre-dip solution **121**, then dry each teat **5** without using towels, and then apply a post-dip solution **127** to each teat **5** after milking System **8** includes a holder in the form of a handpiece **10** connected to a control unit **90** that controls the delivery of high-pressure air **115**, a pre-dip solution **121**, and a post-dip solution **127** to the handpiece **10**.
- (37) As shown in FIGS. **1-5**, the handpiece **10** is a two-part, clamshell structure that forms a hollow handgrip section **12**, a hollow upper body **15**, and a hollow teat receiving body **50**. Mounted on the side of the upper body **15** is a mode switch **21**. Mounted on the front surface of the handgrip section **12** is a trigger button **25**. Mounted on the top surface of the upper body **15** is an optional light **28**. The two half-sections **12A**, **12B** of the handgrip section **12**, the upper body **15**, and the teat receiving body **50** are integrally formed and connected via thread connectors **22**.
- (38) Formed or attached to the handgrip section **12** is a trigger housing **23** containing a trigger switch **26**. Mounted over the trigger switch **26** is a trigger button **25**, hereinafter called a trigger. The upper body **15** includes a mode switch housing **18** and the mode switch **21**. Inside the mode switch housing **18** is a sliding mode button **20** that moves the mode switch **21**. Near the distal end of the proximal end **13** of the handpiece **12** is a conduit end support plate **19**.
- (39) Attached or integrally formed on the upper body **15** is a cylindrical teat receiving body **50**. In the embodiment shown, the teat receiving body **50** is made up of two half-shell structures joined with threaded connectors **22**.
- (40) The teat receiving body includes an outer housing. Formed or attached around the upper end of the outer housing **52** is an upper cap **54**. Formed or attached to the lower end of the outer housing **52** is a lower cap **58**. Formed on the upper cap **54** is a top opening **55** and formed on the lower cap **58** is a lower opening **56**. Located inside the outer housing **52** is an inner cavity **53**.
- (41) Located inside the inner cavity **53** is a cylindrical spray cylinder **62**. FIGS. **9**, **10**, and **11** show that the spray cylinder **62** includes an integrally formed upper section **64** and a lower cavity **76**. Formed inside the spray cylinder **62** is a center bore **63**. Integrally formed on the upper section **64** is a collar-shaped outer housing **65**. Formed inside the outer housing **65** is a closed, circular mixing cavity **66**. Formed on the lower inside surface of the outer housing **65** are air and fluid receiving ports **67A**, **67B** and **67C** that communicate with the mixing cavity **66**. The outer housing **65** and the

mixing cavity **66** extend entirely around the spray cylinder **62**.

(42) The spray cylinder **62** includes a narrow top opening **70** configured to fit under the upper cap **54**. The top opening **55** of the upper cap **54** and the top opening **70** are sufficiently wide to receive the upper portion of a teat **5**. Formed on the lower end of the spray cylinder **62** is a wide bottom opening **74**. During assembly, the wide bottom opening **74** is just inside the lower cap **58**.

(43) As shown in FIG. **10**, the spray cylinder **62** includes a wide upper section **72** near the narrow top opening **70**. Located below the top opening **70** is a plurality of downward aimed nozzles **80**. The nozzles **80** are arranged in a circular pattern on the inside surface of the spray cylinder **62**, so the entire surface of a teat **5** may be treated with high-pressure air **115**, a pre-dip solution **121**, and a post-dip solution **127**. The sidewalls **69** of the spray cylinder **62** below the wide upper section **72** extend inward, forming a narrow neck opening **75** that partially restricts the downward flow of high-pressure air **115**, the pre-dip solution **121**, and the post-dip solution **127** through the spray cylinder **62**.

(44) The lower section **76** of the spray cylinder **62** below the narrow neck opening **75** is a conical shape, diverging and terminates at the spray cylinder's wide bottom opening **59**. The combination of the wide upper section **72**, the downward aimed nozzles **80**, the narrow neck opening **75**, and the diverging, conical-shape lower section **76** all act to create a suction that pulls a teat **5** into the spray cylinder **62** when the teat **5** is positioned $\frac{1}{4}$ to $\frac{1}{3}$ inch above the top opening **70**. Because a teat **5** after milking is flaccid, the suction is also used as a stretching force that causes the teat **5** to stretch longitudinally so that post-dip solution **127** may be applied to the creases and folds commonly found a flaccid teat **5**.

(45) As shown in FIGS. **6-8**, the handpiece **12** includes a set of internal conduits **102**, **104**, and **106** that extend from ports **67A**, **67B**, **67C** formed on a transversely aligned conduit end support plate **19** located in the upper body **15**. The internal conduits **102**, **104**, and **106** extend through the bottom end **16** of the handpiece **12** and connect to external conduits **114**, **120**, and **126**, respectively. The external conduits **114**, **120**, and **126** carry high-pressure air **115**, pre-dip solution **121**, and post-dip solution **127** to the handpiece **12**. During use, high-pressurized air **115**, pre-dip solution **121**, and post-dip solution **121** is delivered via the external conduits **114**, **120**, **126** to the internal conduits **102**, **104**, and **106**, and then to the ports **67A**, **67B**, and **67C**, respectively. The high-pressurized air **115**, pre-dip solution **121**, and post-dip solution **127** is then separately delivered to ports **67A**, **67B**, and **67C**, respectively, formed on the outer housing **65** on the spray cylinder **62**. The high-pressurized air **115**, pre-dip solution **121**, and post-dip solution **127** are then delivered to the circular large mixing cavity **66**. The high-pressure air **115**, pre-dip solution **121**, and post-dip solution **127** then exit the mixing cavity **66** through the nozzles **80** and flow against the teat **5**.

(46) Extending into the handpiece **12** is an electrical cable **100** that contains wires **109**, **110**, and **111** that connect to the light **28**, the mode switch **21**, and the trigger switch **26**, respectively. The end of electrical cable **100** connects to the control unit **90**.

(47) FIGS. **12-19** show another embodiment of the handpiece **12** with a second embodiment of the teat receiving body **200** that can be used with an exchangeable spray cylinder **250**. The teat receiving body **200** is a cylindrical structure designed to be selectively attached to the proximal end **13** of the handpiece **12** with two threaded connectors **22**. The handpiece **12** shown in FIGS. **12-19**, is nearly identical to the embodiment of the handpiece **12** shown in FIGS. **1-6** except that the high-pressure port **67A**, pre-dip solution port **67B**, and post-dip solution port **67C** are aligned vertically on the handpiece's proximal end **13**.

(48) The teat receiving body **200** includes a center bore **202** divided into an upper cylindrical space **262**, and a lower cylindrical space **270**. The air pressure port **222**, the pre-dip port **224**, and the post-dip port **224** formed on the teat receiving body **200** extend inward and communicate with the center bore **202**. The spray cylinder **250** fits and rests inside the center bore **202**. The teat receiving body **200** has a bottom opening **208** and downward diverging sidewalls **216** which create a venturi and produce a suction force that draws the teat into the top opening **204**.

(49) The teat receiving body **200** includes a mounting surface **220** that presses against the proximal end **13** on the handpiece **12**. Formed on the mounting surface **220** is a high-pressure air port **222**, a pre-dip port **224**, and a post-dip port **226**. Ports **222**, **224**, and **226** are vertically aligned and spaced apart so they are aligned and registered with ports **67A**, **67B** and **67C**, respectively, formed on the handpiece **12**. When the teat receiving body **50** is pressed against the proximal end **13** on the handpiece **12**, the ports **222**, **224**, and **226** are configured to slide into ports **67A**, **67B**, and **67C**, respectively. The connectors **22** force the teat receiving body **200** against the proximal end **13**, so that adjacent ports are forced together to create airtight seals between them.

(50) Spray cylinder **250**, shown in FIG. **18**, is a separate structure and is configured to slide into the center bore **202** formed in the teat receiving body **200**. Spray cylinder **250** includes a central bore **252** with a narrow top opening **254** and a bottom opening **255**. The diameter of the central bore **250** is between 20 to 40 mm. The sidewalls **25** of the spray cylinder **250** below the top opening **254** extend inward, forming a narrow neck opening **257** that partially restricts the downward flow of high-pressure air **115**, the pre-dip solution **121**, and the post-dip solution **127** through the spray cylinder **250**.

(51) Like spray cylinder **62**, the lower space **270** on the spray cylinder **250** below the narrow neck opening **254** is a conical shape, diverging and terminates at the spray cylinder's bottom opening **255**. The lower end surface **260** of the spray cylinder **250** rests on a stop edge **218** formed on the inside surface of the teat receiving body **200**.

(52) The spray cylinder **250** is configured to be selectively inserted and removed from the teat receiving body's center bore **202** (shown more clearly in FIG. **14**). The outer surface of the sidewall **256** bends inward and forms at least two circular raceways—an upper raceway **275**, and a lower raceway **278** when the spray cylinder **250** is inserted into the teat receiving body **200**. Three O-rings **280** are attached to the spray cylinder's outer surface above and below the raceways **275**, **278** to isolate the two raceways **275**, **278** from each other and to prevent air and solution leakage.

(53) When inserted into the teat receiving body **200**, the upper raceway **275** is aligned with the high-pressure air port **222** extending into the center bore **202**. The lower raceway **278** is larger than the upper raceway **275** and is aligned both with the pre-dip port **224** and the post-dip port **226**.

(54) Formed on the inside surface of the spray body **250** adjacent to the upper raceway **275** is a first set of high-pressure air nozzles **285**. During use, high-pressure air **115** is delivered via the high-pressure air port **222** on the teat receiving body **200** flows into upper raceway **275** and exits from the first set of high-pressure air nozzles **285**. Pre-dip solution **121** and post-dip solution **127** are delivered to the lower raceway **278** and exit through a second set of nozzles **287** that communicate with the lower cavity **278**.

(55) FIG. **19** is a sectional, side elevational view of another embodiment of the spray cylinder that has an upper raceway **275**, an intermediate raceway **277** and a lower raceway **278** that communicate with the air port **222**, the pre-dip solution port **224**, and the post-dip solution port **226**, respectively. Each raceway **275**, **277** and **278** is separated from an adjacent raceway by o-rings **280** (four total) that prevent the air and solutions delivered to raceways from mixing.

(56) Like the embodiment of the spray cylinder **200** shown in FIG. **19**, the spray cylinder **200** has also has a first set of high-pressure air nozzles **285** adjacent to the upper raceway **222**, a second set of nozzles **286** that communicate with the intermediate raceway **224**, and a third set of nozzles **287** that communicate with the lower raceway **226**. During use, high-pressure air **115** is delivered via the high-pressure air port **222** on the teat receiving body **200** flows into upper raceway **275** and exits the high-pressure air nozzles **285**. Pre-dip solution **121** and post-dip solution **127** are separately delivered to the intermediate and lower raceways **277** and **278** and exit through the second and third set of nozzles **286** and **287**.

(57) FIG. **20** is a sectional, side elevational view of another embodiment of the spray cylinder **200** with smaller diameter central port **202** than the spray cylinder shown in FIG. **19**. The spray cylinder shown in FIG. **20** has two raceways—upper raceway **275** that communicates with an air port **222**

and a lower raceway **278** that communicates with a port designated as a pre-dip solution port **224** or a post-dip solution port **226**. O-rings **280** are placed around the spray cylinder **200** to prevent leakage and prevent mixing of air and solutions in the raceways **275-278**.

(58) In all of the embodiments of the spray cylinders **200**, there are six to eight air nozzles **285** and eight to sixteen liquid nozzles **286, 287**. Some or all of the nozzles **285, 286, and 287** may be aimed downward at 30 to 45 degrees. Some nozzles **285, 286, 287** aimed radially thereby forcing the air and solutions to travel in a circular path along the inside surface of the spray cylinder **230**. It should also be noted that the type and number of nozzles **285, 286, 287** formed on the spray cylinders **250** may also vary. Because the control unit **90** is programmable by the manufacturer, the control unit **90** may include optional mode settings that allow a user to use different spray cylinders (not shown) with a different arrangement of nozzles for different operational modes.

(59) Also, as shown in FIGS. **12-15, and 17**, a deflection cap **300** may be integrally formed or attached to the modified spray cylinder **250** lower edge. The deflection cap **300** is configured to re-focus and concentrate the high-pressure air, pre-dip solution and post-dip solutions exiting the lower section of the teat receiving body into a narrow stream. The deflection cap **330** has a complex curved outer body that curves inward and forms a point. The air, pre-dip, and post-dip solutions stream is narrow and redirected approximately 20 to 45 degrees from the horizontal axis. By narrowing and re-focusing the stream inward, overspray is prevented. While holding the handpiece, the user may aim the high-pressure air **115**, pre-dip solution **121**, and/or post-dip solution **127** at excessively dirty or hard-to-reach areas on the teat or udder. It should be understood that the deflection cap **112** may also be integrally formed or attached to the handpiece shown in FIGS. **1-8**.

(60) FIGS. **21 and 22** are views of the teat receiving body **200** containing a spray cylinder **250** and the deflection cap **300** with a diffuser **302** and agitator **305** positioned transversely inside the deflection cap **300**. During operation when with high pressure air and pre-dip and post dip solutions flow through the deflector cap **300**, the diffuser **302** and agitator **303** disturb the linear flow of the downward flow of air, and the solutions, which creates turbulence and increase their distribution inside the deflection and through the deflection cap's end opening.

(61) In all of the above embodiments described above, the handpiece **12** is a part of a system **8** that includes a control unit **90**. The control unit **90** includes programmable logic control, called a PLC indicated by reference number **92**, with a software program **95** loaded into its working memory. The PLC **92** and the software program **95** are configured to control the delivery of high-pressure air precisely **115**, pre-dip solution **121**, and post-dip solution **127** to the handpiece **12**. As shown in FIG. **18**, the control unit **90** is connected to a first solenoid **112**, a second solenoid **118**, and a third solenoid **124**. The first solenoid **112** is connected to a pressurized air source **113** and to an external conduit **114** that connects to the internal conduit **102** inside the handpiece **12**. The second solenoid **118** is connected to a pre-dip solution source **119**, and the external conduit **120** connects to an internal conduit **102** inside the handpiece **12**. The pre-dip solution **121** may be delivered to an optional heater **128** connected to the conduit **120** that pre-heats the pre-dip solution **121** to approximately 90 to 120 degrees F. before delivery to the internal conduit **102**.

(62) Third solenoid **124** is connected to a post-dip solution source **125** and an external conduit **126** that connects to internal conduit **106** inside the handpiece **10**. The third solenoid **124** may also be connected to an optional heater **129** connected to the conduit **126** that pre-heats the post-dip solution **127** to approximately 90 to 120 degrees F. before delivery to the nozzles **80**.

(63) As discussed further below, the software program **95** is configured to control when and how long each solenoid **112, 118, and 124** opens and closes. During use, the operator manipulates the mode switch **21** to select one of two operating modes—a pre-dip solution application mode: and a post-dip solution application mode. The two modes determine which solenoid opens and closes and how long. The operator then manipulates trigger **25** to initiate the operating mode selected by the mode switch **21**.

Operation of the Cleaning, Pre-Dip Application, and Post-Dip Solution Embodiment

(64) When the mode switch **21** is moved to the first mode position, and the operator presses the trigger **25**, the PCL **92** is activated, which runs the first stage mode routine. During the first mode routine, the software program **95** and the PLC **92** open the first solenoid **112** connected to the pressure air source **113** containing high-pressure air **115** (approximately 150 PSI). During the first mode routine, the second solenoid **118** is opened simultaneously with the first solenoid **112**. When the second solenoid **118** is opened, pre-dip solution **121** is delivered to the handpiece **10**. When the second solenoid **118** is activated, the optional heater **129** may also be activated to heat the pre-dip solution **121**.

(65) The software program **95** is configured to open both the first and second solenoids **112**, **118**, respectively, when trigger **25** is continuously pressed. Both high-pressure air **115** and the pre-dip mixture **121** are delivered to handpiece **12** and discharged from the nozzles **80**. The first and second solenoids **112**, **118** remain open as long as trigger **25** is pressed.

(66) When trigger **25** is released, the software program **95** is configured to close the second solenoid **118** immediately, thereby discontinuing the pre-dip solution **121** to nozzles **80**. In addition, the software program **95** controls the PLC **92** so that the first solenoid **112** remains open for approximately 2 seconds after the second solenoid **118** closes, allowing only high-pressure air **115** to flow from the nozzles **80** and remove excessive pre-dip solution **121** from the surface of the teat **5**. If the operator presses the trigger **25** again while the first solenoid **112** is opened and the second solenoid **118** is closed, the first stage mode routine automatically re-runs.

(67) When the mode switch **21** is moved to the post-dip application mode position and the trigger **25** is pressed, the PCL **92** runs the second stage mode routine. In the second stage mode routine, the first solenoid **112** opens to deliver high-pressure air **115** to the nozzles **80**. The first solenoid **112** is open as long as the trigger **25** is pressed. When trigger **25** is released, the third solenoid **124** opens to deliver post-dip solution **127**. If the third solenoid **124** is connected to an optional heater **129**, the post-dip solution **127** is heated before delivery to the handpiece **12**.

(68) When trigger **25** is released, the first solenoid **112** remains open for 0.25 to 0.5 seconds, allowing high-pressure air **115** to be released from the nozzles **80** and mixed with the post-dip solution **127**. The mixture of high-pressure air **115** and post-dip solution **127** causes the post-dip solution **127** to atomize and spread across the entire surface of the teat **5**. The software program **95** is also configured to keep the third solenoid **124** open 0.25 to 0.5 seconds after the first solenoid **112** closes so that the entire teat **5** is wetted with post-dip solution **127**.

Stripping Embodiment

(69) FIGS. **19-21** show another embodiment of system **8** that includes the handpiece **12** and control unit **90** used in the first embodiment discussed above. In this embodiment, the software program is re-programmed or replaced by a modified software program **96**, and the spray cylinder **62** is replaced by a stripping spray cylinder **130**. The stripping spray cylinder **130** includes the same upper section **64** with an outer housing **65**, a mixing cavity **66**, and nozzles **80**. During use, a teat **5** is placed over the top opening of the upper section **64** and drawn into the center bore. High-pressure air **115**, pre-dip solution **121**, and post-dip solution **127** are then delivered to and dispensed from the nozzles **80** described above in the spray cylinder **62**, **250**.

(70) The stripping spray cylinder **130** is configured to rest over a cylindrical, inflating member **140** placed inside the teat receiving body **50**. The inflating member **140** is stacked above a short conical support structure **180**.

(71) The inflating member **140** includes a cylindrical bladder frame **150** that fits into the teat receiving body **50**. As shown in FIG. **23**, the bladder frame **150** includes a top opening **151**, a bottom opening **152**, and sidewalls **153** that converge in a downward direction forming a conical-shaped center passageway **155**. Extending through a sidewall **153** is a laterally extending air inlet **154** that communicates with a center passageway **155** formed in the bladder frame **150**. The air inlet **154** is located near the middle section of the bladder frame **150**. Also formed on the bladder

frame **150** is a small exhaust port **171** located on the opposite side of the bladder frame **105** that also communicates with the inner air cavity **170**. Extending along the inside surface of the sidewall **152** and aligned with the exhaust port **171** formed on the bladder frame **150** is a circular channel **174**.

(72) As shown in FIG. 23, the bladder frame **150** rests on top of a short conical support structure **180** that fits inside the teat receiving body **50**. The conical support structure **180** includes a top opening **182**, and a bottom opening **184**. Formed around the top opening **182** is a wide-diameter upper lip **188**. Formed around the inside diameter of the short conical support **180** are diverging sidewalls **186**.

(73) Located inside the bladder frame **150** is an elastic bladder sleeve **160** is configured to sequentially expand and relax repeatedly against the sides of a teat **5**. The bladder sleeve **160** is a ballooning, cylindrical-shaped structure with thin upper sidewalls **164** (approx. 1 ml thick) relatively thick lower sidewalls **166** (approximately 3 to 4 mils thick). The bladder sleeve **160** includes a circular upper edge **162** configured to stretch over the upper edge of the bladder frame **150** and a lower edge **164** configured to stretch over the lower edge of the bladder frame **150**. The upper and lower edges **162**, **164** affix the bladder sleeve **160** to the bladder frame **150**. The inside surface of the sidewalls of the bladder sleeve **160** extends inward and narrows from top to bottom and then extends outward to form an adjustable, hourglass-shaped center passageway **166** in which the teat **5** is placed.

(74) When the bladder sleeve **160** is assembled on the bladder frame **150**, the center, outside sidewalls of the bladder sleeve **160** are detached from the bladder frame **150**. An inner air space **170** is formed between the inside surface of the bladder frame **150** and the outside surface of the bladder sleeve **160**. The air space **170** is partially closed so that air trapped inside the inner air space **170** escapes only through the exhaust port **171**. Because the center passageway **166** has an hour-glass configuration and the thickness of the sidewall **164** of the bladder sleeve **160** is thinner in the upper portion and therefore more flexible than the lower portion of the bladder sleeve **160**, the upper portion of the bladder sleeve **160** expands inward before the lower portion to gradually squeeze the teat **5** in a downward, wave-like manner. Because the upper portion of the bladder sleeve **160** is relatively thin (1 to 2 mls), low-pressure air **116** (approximately 50 PSI) must expand the bladder sleeve **160**.

(75) During use, delivery of the low-pressure air **116** to the inner air-spaced **170** is repeated, continued, and discontinued (called inflation/deflation cycles) to create a plurality of squeezing, milking-like movements against the teat **5**. When delivery of the low-pressure air **116** is discontinued, the low-pressure air **116** in the inner air space **170** slowly escapes from the inner air space **170** via the exhaust port **171**. Because the circular recessed channel **174** is formed in the inside surface of the bladder frame **150** that communicates with the exhaust hole **172**, any air blocked by the collapsing bladder sleeve **160** may escape through the exhaust hole **172**.

(76) Delivery of the low-pressure air **116** is controlled by the PCL **92** and the modified software program **96**. As shown in FIG. 19, the control unit **90** is connected to a fourth solenoid **132** that is connected to a low-pressure air source **135** (approximately 50 PSI). It should be understood that the fourth solenoid **132** may be connected to high-pressure air source **113** and configured to lower the air pressure from 150 PSI to approximately 50 PSI. A conduit **133** runs from the fourth solenoid **132** to the handpiece **12** and the air outlet conduit **156** inside the handpiece **12** and then to the bladder frame **150**.

Operation of Stripping Embodiment

(77) The washing stage and drying stage used in this embodiment is identical to the washing and drying stages used in the first two embodiments shown in FIG. 1-18. The only difference is that in the third embodiment, is the ability to strip the teat **5** after applying the pre-dip solution **121**.

(78) Immediately after the drying step in the pre-dip application stage is completed, the software program **95** and PCL **92** automatically begin a stripping mode that involves opening and closing the

fourth solenoid **132** connected to the lower pressure source **135** or the high-pressurized air source **113**. The fourth solenoid **132** connects to an external conduit **133**, connecting to an air outlet conduit **156** inside the handpiece **12**. The air outlet conduit **156** connects to the inner air space **170**. Air escapes from the inner air space **170** via the exhaust port **171**.

(79) The cyclic, ON/OFF operation of the fourth solenoid **132** is controlled by the PCL **90** and the software program **95**. During the stripping stage, 3 to 4 inflation and deflation cycles are performed. Each cycle delivers low pressurized air **116** to the inner air space **170** for approximately 0.5 seconds, followed by a discontinuation period of approximately 0.5 seconds, which enables the low pressurized air **116** to escape. After the stripping step has been completed, the operator then moves the mode switch **21** to the second mode operation position. When the operator then presses the trigger **25**, the PCL **92** is activated and runs the post-dip application mode routine, which opens the first solenoid **112** (high-pressure air). When trigger **25** is released, the third solenoid **124** opens to deliver post-dip solution **127** to the nozzles **80**. When the trigger **25** is released, the first solenoid **112** remains open for 0.25 to (0.5) seconds, allowing high-pressure air **115** to be released from the nozzles **80** and mixed with the post-dip solution **127** and atomize the last volume of post-dip solution **127** exiting the nozzles **80**. The third solenoid **124** remains open 0.25 to 0.5 sec after the first solenoid **112** closes so the entire teat **5** is wetted with a layer of post-dip solution **127**.

(80) In summary, the above system **8** uses high-pressure air delivered to a handpiece **12** to apply a pre-dip solution **121** and a post-dip solution **127** to a teat **5**. The handpiece **12** includes a spray cylinder **62** or a stripping spray cylinder **130** with nozzles **80** to evenly spray high-pressure air **115**, the pre-dip solution **121**, and post-dip solution **127** around the teat **5**. Below the spray cylinder, **62** or the stripping spray cylinder **130** is a venturi-generated device that creates a vacuum that draws and stretches the teat **5** inside the teat receiving body. System **8** can also be easily modified to include a stripping feature that strips the teat **5** before applying the post-dip solution **127**.

Robotic System Embodiment

(81) The above system described system uses a handpiece that allows for manual teat preparation. However, the above system can be modified for robotic implementation. For robotic implementation, the handpiece is removed and replaced by a robotic arm system **400** that requires the use of the teat receiving body **200** containing of one of the spray cylinders **250** described above. The teat receiving body **200** is connected to a control unit that controls the sequential flow of high-pressure air, pre-dip solution and post-dip solution to the teat receiving body **200**.

(82) The robotic arm system **400**, shown in FIGS. **31-35**, includes a rotating support arm **410** attached to a motor **414**. The motor **414** is mounted to a rotating platform **418**. The motor and support arm **410** are configured so that support arm **410** rotates in a vertical arc of approximately 180 degrees. The rotating platform **418** is configured to rotate in a horizontal arc 180 to 360 degrees.

(83) Mounted to the distal end of the rotating support arm **410** is a second support arm **402** configured to rotate in a horizontal arc of 180 to 360 degrees. Mounted on the distal end of the second support arm **402** is a pivoting arm **406** configured to rotate from the distal end of the second support arm **402** in a vertical arc of 180 degrees. Mounted on the outer end surface of the pivoting arm **406** are two teat receiving bodies **200**.

(84) FIG. **32** is a perspective view of a robotic head assembly **400** with two teat receiving bodies attached to opposite sides of a single support arm **406**.

(85) FIG. **33** is a perspective view of a robotic head assembly **400** with four teat receiving bodies **200** attached to a single support arm **406**.

(86) FIG. **34** is a partial perspective of the second support arm **402** used with the robotic head assembly also uses four teat receiving bodies **200** attached to two moveable mounting arms **460** that are perpendicularly aligned to a transversely oriented alignment arm **450**. Each alignment arm **450** is mounted on the end of a short support arm **406**. The mounting arms **460** are perpendicularly aligned and extend outward from the alignment arm **450**. The mounting arms **460** are also

moveable along the alignment arm **450**, enabling the system to adjust the spacing between the teat receiving bodies **200**. Each teat receiving body **200** is attached to two rods **470** that connect to a mounting arm **460**. FIG. **34** also shows the mounting arms **460** in a narrow configuration and FIG. **35** shows the mounting arms **460** in a wide configuration.

(87) FIG. **36** is a perspective view of the robotic arm system **400** mounted on track **500** with a track control unit **510** connected to track **500**. The track control unit **510** controls the movement of the robotic arm system **400** along the track, the movement of the platform **418**, arms **402**, **406**, **410**, **450**, and **460**. The control unit **510** may be connected with the main control unit **90** which controls the delivery of the high-pressure air, pre-dip solution and post dip solution.

(88) In compliance with the statute, the invention described has been described as more or less specific to structural features. It should be understood, however, that the invention is not limited to the specific features shown, since the means and construction is shown, comprises the preferred embodiments for putting the invention into effect. The invention is therefore, claimed in its forms or modifications within the legitimate and valid scope of the amended claims, appropriately interpreted under the doctrine of equivalents.

Claims

1. A teat preparation apparatus, comprising: a. a holder; b. a teat receiving body mounted to said holder, said teat receiving body including sidewalls that form a center bore, and a high-pressure air port that connects to a high-pressure air conduit, a pre-dip port that connects to a pre-dip conduit and a post-dip port that connects to a post-dip conduit, said high-pressure air port being located above said pre-dip port and said post-dip port; c. a spray cylinder located inside said center bore in said teat receiving body, said spray cylinder including sidewalls that form a central bore configured to fit around a teat, said sidewalls of said spray cylinder configured to form an upper raceway that communicates with said high-pressure air port, said sidewalls also configured to form at least one lower raceway located below and isolated from said upper raceway that communicates with said pre-dip port or said post-dip port; d. a first set of nozzles formed on said spray cylinder adjacent to said upper raceway configured to deliver said high-pressure air to said central bore; and e. a second set of nozzles formed on said spray cylinder adjacent to said lower raceway configured to deliver said pre-dip solution or said post-dip solution to said central bore.
2. The teat preparation apparatus, as recited in claim 1, further including a plurality of o-rings disposed around said spray cylinder, at least one said o-ring being located between said upper raceway and said lower raceway to prevent high-pressure air from flowing into said lower raceway and to prevent said pre-dip solution or said post-dip solution delivered to said lower raceway from flowing into said upper raceway.
3. The teat preparation apparatus, as recited in claim 2, wherein said holder is a handpiece connected to said teat receiving body.
4. The teat preparation apparatus, as recited in claim 2, further including a deflector cap extending downward from said teat receiving body.
5. The teat preparation apparatus, as recited in claim 4, further including a diffuser inside said deflector cap.
6. The teat preparation apparatus, as recited in claim 5, further including an agitator attached to said diffuser configured to disrupt the downward flow of said high-pressure air, said pre-dip solution or said post-dip solution through said deflector cap.
7. The teat preparation apparatus, as recited in claim 1, wherein said second set of nozzles is angled upward.
8. The teat preparation apparatus, as recited in claim 1, further including an intermediate raceway being located between said upper raceway and said lower raceway, said intermediate raceway communicating with said pre-dip port or said post-dip port.

9. The teat preparation apparatus, as recited in claim 8, further including a third set of nozzles located adjacent to said intermediate raceway configured to deliver said pre-dip solution or said post-dip solution to said central bore.
 10. The teat preparation apparatus, as recited in claim 9, wherein said holder is a handpiece connected to said teat receiving body.
 11. The teat preparation apparatus, as recited in claim 8, further including at plurality of o-ring disposed around said spray cylinder, at least one said o-ring located between said upper raceway and said second raceway and at least one second o-ring located between said lower raceway and said intermediate raceway.
 12. The teat preparation apparatus, as recited in claim 8, further including a control unit connected to a high-pressure air source, to a pre-dip solution source, and to a post-dip solution source, said control unit configured to control the flow of high-pressure air from said high-pressure air source to said high-pressure port, and control the flow of pre-dip solution from said pre-dip post-dip solution source to said pre-dip port, and control the flow of post-dip solution from said post-dip source to said post-dip port.
 13. The teat preparation apparatus, as recited in claim 1, further including a deflector cap extending downward from said teat receiving body.
 14. The teat preparation apparatus, as recited in claim 13, further including a diffuser inside said deflector cap.
 15. The teat preparation apparatus, as recited in claim 14, further including an agitator attached to said diffuser configured to disrupt the downward flow of said high-pressure air, said pre-dip solution or said post-dip solution through said deflector cap.
 16. The teat preparation apparatus, as recited in claim 1, further including a control unit connected to a high-pressure air source, to a pre-dip solution source, and to a post-dip solution source, said control unit configured to control the flow of high-pressure air from said high-pressure air source to said high-pressure port, and control the flow of pre-dip solution from said pre-dip post-dip solution source to said pre-dip port, and control the flow of post-dip solution from said post-dip source to said post-dip port.
 17. The teat preparation apparatus, as recited in claim 1, wherein said holder is a robotic arm control apparatus that selectively adjusts the positioning of said spray cylinder so that said central bore is positioned around a teat.
 18. The teat preparation apparatus, as recited in claim 17, wherein said robotic arm control apparatus is mounted on a track and selectively moves over said track.
 19. The teat preparation apparatus, as recited in claim 1, wherein said teat receiving body is mounted on a support member attached to said robotic arm control apparatus.
 20. The teat preparation apparatus, as recited in claim 19, further including a second teat receiving body mounted on said support member.
 21. The teat preparation apparatus, as recited in claim 20, wherein said teat receiving body and a second teat receiving body are adjustable to simultaneously fit around two teats on an utter.
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